Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.02	20.18	
worker nodes in service, mean	6.00	6.00	-0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.00	2.02	+0.8%
power (W), mean	628	642	+2.3%
power (W), peak	652	688	+5.5%
energy (Wh)	209	216	+3.2%
energy, parked workers still on at idle power (Wh)	209	216	+3.2%
CPU used (cores), mean	0.860	1.24	+44.3%
CPU allocatable (cores), mean	24.00	24.00	-0.0%
utilisation (used / allocatable)	0.036	0.052	+44.3%
energy per core-hour (Wh)	730	518	-29.1%
node-hours per core-hour	6.98	4.84	-30.7%
pending pods, pod-minutes	0.250	0.533	+113.3%
pending pods, peak	1.00	2.00	+100.0%
HPA replicas, mean	9.04	9.43	+4.3%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.767	1.07	+39.1%
requests timed	1,070	1,100	+2.8%
time over the response line (% of samples)	0.187	0.000	-100.0%
response time (ms), mean	125	95.08	-24.0%
response time (ms), median	111	81.30	-26.9%
response time (ms), 95th percentile	249	155	-38.0%
response time (ms), 99th percentile	385	180	-53.3%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.6251	5.3930	-29.3%	[-3.8610, -0.6918]	GAIN
kwh_per_core_hour	0.7945	0.5734	-27.8%	[-0.3766, -0.0673]	GAIN
utilisation	0.0354	0.0508	+43.7%	[+0.0038, +0.0284]	GAIN
pending_pod_minutes_per_hour	0.7200	1.5238	+111.6%	[-2.1600, +4.5714]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.2142	3.0537	+37.9%	[-3.5052, +6.1309]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.359 0.393 0.421 0.446 0.468 0.488 0.504 0.52 0.534 0.545 0.555 0.561 0.566 0.571 0.576 0.581 0.584 0.587 0.589 0.59
- nodes decided per minute: 6 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	614	636	5.00	5.00	0.000	0.500	0.492	1.06

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	618	624	5.25	8.00	0.250	0.000	0.603	0.746
2	6.00	6.00	624	623	6.00	8.00	0.000	0.000	0.758	0.698
3	6.00	6.00	620	634	6.00	8.00	0.000	0.000	0.640	1.01
4	6.00	6.00	634	647	9.25	10.00	0.000	0.000	1.00	1.37
5	6.00	6.00	631	646	10.00	10.00	0.000	0.000	0.965	1.37
6	6.00	6.00	627	646	10.00	10.00	0.000	0.000	0.867	1.36
7	6.00	6.00	643	669	10.00	10.00	0.000	0.000	1.25	1.95
8	6.00	6.00	645	679	10.00	10.00	0.000	0.000	1.32	2.24
9	6.00	6.00	647	670	10.00	10.00	0.000	0.000	1.38	2.00
10	6.00	6.00	640	676	10.00	10.00	0.000	0.000	1.15	2.15
11	6.00	6.00	618	623	10.00	10.00	0.000	0.000	0.588	0.743
12	6.00	6.00	618	624	10.00	10.00	0.000	0.000	0.607	0.755
13	6.00	6.00	620	638	10.00	10.00	0.000	0.000	0.660	0.930
14	6.00	6.00	634	647	10.00	10.00	0.000	0.000	1.03	1.38
15	6.00	6.00	631	646	10.00	10.00	0.000	0.000	0.940	1.38
16	6.00	6.00	631	644	10.00	10.00	0.000	0.000	0.930	1.29
17	6.00	6.00	623	635	10.00	10.00	0.000	0.000	0.724	1.07
18	6.00	6.00	616	624	10.00	10.00	0.000	0.000	0.569	0.705
19	6.00	6.00	618	627	10.00	10.00	0.000	0.000	0.593	0.822

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.